

1. An airplane flies horizontally from east to west at 320 mi/hr relative to the air. If it flies in a steady 40-mi/hr wind that blows horizontally toward the southwest (45° south of west), find the speed and direction of the airplane relative to the air.

2. Which has a greater horizontal component, a 100-N force directed at an angle of 60° above the horizontal or a 60-N force directed at an angle of 30° above the horizontal?

3. In still air, a parachute with payload would fall vertically at a terminal speed of 40 m/s. Find the direction and magnitude of its terminal velocity relative to the ground if it falls in a steady wind blowing horizontally from west to east at 10 m/s.

4. A woman in a kayak paddles due west at 4 mi/hr relative to the water in a current that flows northwest at 2 mi/hr. Find the speed and direction of the canoe relative to the shore.

5. If a 500-lb load is suspended from the ceiling by two chains such that the angle between each chain and the ceiling is 45° , what is the magnitude of the force each chain must be able to withstand?